

Decomposition.

$$\begin{bmatrix} y_{1,it} \\ y_{2,it} \end{bmatrix} = \begin{bmatrix} \mu_{1,i} \\ \mu_{2,i} \end{bmatrix} + \begin{bmatrix} y_{1,it}^w \\ y_{2,it}^w \end{bmatrix}$$

Within-level model.

$$\begin{bmatrix} y_{1,it}^w \\ y_{2,it}^w \end{bmatrix} = \begin{bmatrix} \phi_{(1)11,i} & \phi_{(1)12,i} \\ \phi_{(1)21,i} & \phi_{(1)22,i} \end{bmatrix} \begin{bmatrix} y_{1,i(t-1)}^w \\ y_{2,i(t-1)}^w \end{bmatrix} + \begin{bmatrix} \zeta_{1,it} \\ \zeta_{2,it} \end{bmatrix},$$

with $\zeta_{k,it} \sim N(\mathbf{0}, \sigma_{\zeta_k,i}^2)$

Between-level model.

$$\begin{bmatrix} \mu_{1,i} \\ \mu_{2,i} \\ \phi_{(1)11,i} \\ \phi_{(1)12,i} \\ \phi_{(1)21,i} \\ \phi_{(1)22,i} \\ \ln(\sigma_{\zeta_1,i}^2) \\ \ln(\sigma_{\zeta_2,i}^2) \end{bmatrix} = \begin{bmatrix} \gamma_{0,\mu_1} \\ \gamma_{0,\mu_2} \\ \gamma_{0,\phi_{(1)11}} \\ \gamma_{0,\phi_{(1)12}} \\ \gamma_{0,\phi_{(1)21}} \\ \gamma_{0,\phi_{(1)22}} \\ \gamma_{0,\ln(\sigma_{\zeta_1}^2)} \\ \gamma_{0,\ln(\sigma_{\zeta_2}^2)} \end{bmatrix} + \begin{bmatrix} v_{\mu_1,i} \\ v_{\mu_2,i} \\ v_{\phi_{(1)11,i}} \\ v_{\phi_{(1)12,i}} \\ v_{\phi_{(1)21,i}} \\ v_{\phi_{(1)22,i}} \\ v_{\ln(\sigma_{\zeta_1}^2),i} \\ v_{\ln(\sigma_{\zeta_2}^2),i} \end{bmatrix},$$

with $v_i \sim MVN(\mathbf{0}, \mathbf{\Omega})$